

The exam paper includes 15 pages

▪ Exam instructions:

- All MCQs have to be answered in bubble sheet.
- Make sure that your exam form is similar to bubble sheet model.
- Write answers of MCQ in bubble sheet by blue pen only.
- Double answer of any MCQ will be considered wrong.

1) MCQ (select the best answer) (60 questions, 0.5 mark for each question)

1- Which one of the following is not true regarding SCLC?

- ☒ A. Surgery is the modality of choice in limited stage of SCLC.
- ☐ B. Regardless of stage, chemotherapy is an essential part of treatment.
- ☐ C. Radiation treatment may be used as adjuvant depending on the stage of cancer.
- ☐ D. Para neoplastic syndrome is common association.

2- Which of the following is the most common cell type in scar carcinoma?

- ☐ A. Small cell carcinoma.
- ☒ B. Squamous cell carcinoma.
- ☐ C. Adenocarcinoma.
- ☐ D. Large cell carcinoma.

3- Which one of the following statements is true of mesothelioma?

- ☒ A. Incidence is decreasing, reflecting the ban on use of asbestos.
- ☐ B. The disease is more common in females.
- ☐ C. All deaths due to mesothelioma should be discussed with the coroner.
- ☐ D. Radical surgery is the treatment of choice.

4- You are asked to interpret an arterial blood gas of a 76-year-old patient who admitted to accident and emergency with an acute onset of breathlessness and low oxygen saturations. The test was taken on room air and read as follows: pH= 7.37, PO₂=58 mmHg, PCO₂=31 mmHg, HCO₃= 24, SO₂=89%. Choose the most likely interpretation for this arterial blood gases:

- ☐ A. Compensated respiratory acidosis.
- ☒ B. Type 1 respiratory failure.
- ☐ C. Uncompensated respiratory alkalosis.
- ☐ D. Type 2 respiratory failure.

5- Typical symptoms and signs caused by a high PaCO₂ include:

- ☐ A. Cold pale skin.
- ☐ B. Weak pulse.
- ☒ C. Headache, which is often worse following sleep.
- ☐ D. Central cyanosis.

6- A 20-year-old, female, presented with paroxysmal attacks of dyspnea and chest wheezes. Her chest X-ray detected no abnormalities but her spirometry tests revealed normal FVC, low FEV₁ and FEV₁/FVC. The most probable diagnosis is

- ☐ A. Restrictive lung disease.
- ☒ B. Obstructive lung disease.
- ☐ C. Chest wall disease.
- ☐ D. Vascular lung disease.

7- Which of the following is most frequently observed in the posterior mediastinum?

- ☒ A. Neurogenic tumors.
- ☐ B. Mesothelioma.
- ☐ C. thymomas .
- ☐ D. cholangiocarcinoma.

8- The following would not be typical for the presentation of a pleural effusion:

- A. Dyspnoea.
- B. Chest pain.
- C. Reduced effort tolerance.
- ☒ D. Nausea and Vomiting.

9- Which of the following structures forming the right border of the heart?

- A. Left auricle
- B. Ascending aorta
- ☒ C. Right atrium
- D. Right ventricle

10- Which of the following chambers are separated by the coronary sulcus?

- A. Right atrium and left ventricle
- ☒ B. Right atrium and right ventricle
- C. Right and left atria
- D. Right and left ventricles

11- A 13-year-old girl presents with a complaint of fatigue and shortness of breath with moderate exercise. She has no significant past medical history. Her HR is 110/min, RR 20/min, BP 115/65 mmHg, and T 36.9 °C. On physical exam, her second heart sound is found to be widely split and fixed in all phases of respiration. A systolic ejection murmur is heard at the left middle and upper sternal border. Her chest x-ray shows enlargement of the right atrium and ventricle. Which of the following is the most likely cause of her murmur?

- A. Turbulent flow across an ASD
- B. Turbulent flow across a VSD
- C. Increased flow across the aortic valve
- ☒ D. Increased flow across the pulmonary valve

12- A 15-month-old male presents to the emergency department with weakness, poor feeding, failure to thrive, tachypnea, and a nonproductive cough.. He has had no fevers or sick contacts. Physical exam reveals a HR of 142/min, BP 95/45 mmHg, RR 60/min, and T 36.8 °C, and SpO2 is 94%. He is awake and alert and tachypneic with mild wheezing and retractions. He has marked hypotonia, almond-shaped eyes with epicanthal folds, and a large protruding tongue. He has an active precordium and a pulmonary ejection murmur as well as a holosystolic murmur at the apex radiating to the axilla. His liver is enlarged. Pulses are equal on all four extremities. His chest x-ray shows cardiomegaly and increased pulmonary vascular markings.. Which one of the following is the most appropriate therapy?

- A. Inhaled albuterol
- B. Inhaled nitric oxide
- ☒ C. Furosemide
- D. Supplemental oxygen

13- Which one of the following life-threatening congenital anomalies in the newborn presents with polyhydramnios, aspiration pneumonia, excessive salivation, and difficulty in passing a nasogastric tube

- A. Gastroschisis
- B. Diaphragmatic hernia
- ☒ C. Tracheoesophageal fistula
- D. Choanal atresia

14- Which of the following is suggestive of extrapulmonary TB

- A. Failure to thrive
- B. Bronchial breath sound
- ☒ C. Lymphadenopathy
- D. Wheezing

15- Horner's syndrome is a complication of which of the following lesion?

- A. Bronchial carcinoid
- ☒ B. Right apical carcinoma
- C. Apical lung abscess
- D. Right basal carcinoma

16- Which of the followings lung tumor has a tendency to undergo necrosis and cavitations?

- A. Adenocarcinoma
- B. Bronchioalveolar carcinoma
- C. Carcinoid
- ☒ D. Squamous cell carcinoma

17- Which of the following statements is true about cardiac edema ?

- A. It occurs only in the lower limbs
- ☒ B. It is due to congestive heart failure
- C. It is non pitting edema
- D. Edema fluid is composed of inflammatory exudate

18- Which of the following is true regarding pulmonary circulation?

- ☒ A. It is a low-pressure system that allows for improved gas exchange.
- B. It is the larger of the two circulatory systems.
- C. The system functions with an increased arterial pressure to circulate through the distal parts of the body.
- D. It consists of the left side of the heart, the aorta, and its branches.

19- Why the pulmonary artery BP is lower than the systemic arterial BP ?

- A. The output of the right ventricle is less than that of the left ventricle.
- B. The left ventricular wall is thicker than that of the right ventricle.
- C. The aorta is more elastic than the pulmonary artery.
- ☒ D. The pulmonary resistance is less than the systemic resistance.

20- What is the cardiac ejection fraction?

- A. It is the percentage ratio of cardiac work to total energy expenditure.
- B. It is the cardiac output per square meter body surface area.
- ☒ C. It is the percentage of blood leaving the heart in each cardiac cycle.
- D. It is a premature contraction of the heart occurring during diastole.

21- Unequal upper extremity arterial blood pressure is commonly found in each of the following disorders except:

- A. Aortic dissection.
- ☒ B. Aortic regurgitation.
- C. Tumour compressing subclavian artery.
- D. Arterial Embolism.

22- Canon a-wave caused by synchronous atrial and ventricular contraction is usually found in one of the following disorders :

- A. Atrial fibrillation.
- ☒ B. Complete heart block.
- C. Tricuspid valve disease.
- D. Restrictive cardiomyopathy.

23- Which of the following not a causes of cardiac syncope ?

- A. Adams stoke attacks.
- B. Severe aortic stenosis.
- C. Hypertrophic cardiomyopathy.
- ☒ D. Mitral regurgitation.

24- Pathological Q wave in lead V1-- V4 in resting ECG means:

- ☒ A- Old antroseptal MI.
- B- Old lateral MI.
- C- Old inferior MI.
- D- Old extensive anterior MI.

25- In exercise ECG transient ST elevation reflect?

- ☒ A. Subendocardial ischemia.
- ☒ B. Transmural ischemia.
- C. Epicardial ischemia.
- D. Epicardial infarction

26- Ambulatory BP monitoring interpretation includes:

- A. Arrhythmia detection
- ☒ B. Nocturnal dipping
- C. Assessment of myocardial ischemia
- D. Evaluation of response to antiarrhythmic medications

27- Radio femoral delay is the cardinal signs in patients with:

- A. Aortic aneurysm
- B. Aortic dissection
- ☒ C. Coarctation of the aorta
- D. Mitral stenosis

28- Stanford& DeBakey classifications are used to classify:

- A. Peripheral vascular disease
- ☒ B. Aortic dissection
- C. Renal artery stenosis
- D. Internal carotid artery lesions

29- Which of the following statements is true about reentry mechanism of cardiac arrhythmia?

- ☒ A. It requires 2 pathways of different refractoriness and conduction with unidirectional block
- B. It represents 10% of cases of arrhythmias
- C. It requires 2 pathways of the same conduction with bidirectional block
- D. It has gradual onset and offset

30- What is the most common cause of ventricular tachycardia (VT)?

- ☒ A. Myocardial ischemia
- B. Electrolytes imbalance
- C. Renal failure
- D. Interstitial pulmonary fibrosis

31- The chamber marked by an arrow in the following 2D echocardiography is:

- ☒ A. Left ventricle
- ☐ B. Mitral valve
- ☐ C. Left atrium
- ☐ D. Right ventricle



32- Which of the following is not part of atherosclerotic plaque?

- ☐ A. Fibrous cap
- ☐ B. Cholesterol accumulation
- ☐ C. Lipid core.
- ☒ D. Platelets.

33- The main difference between unstable angina and NSTEMI?

- ☐ A. The frequency of chest pain
- ☐ B. The duration of chest pain
- ☒ C. Cardiac enzyme elevation
- ☐ D. ECG criteria

34- Which valves are commonly affected by Rheumatic fever:

- ☐ A. Tricuspid , mitral
- ☒ B. Mitral , Aortic
- ☐ C. Tricuspid , pulmonary
- ☐ D. Aortic , pulmonary

35- A 13 year old female patient with rheumatic fever comes with involuntary dancing movement; how can you explain this movement:

- ☐ A. Myopathy
- ☐ B. Neuromuscular junction disease
- ☒ C. Sydenham's chorea
- ☐ D. Psychosis

36- Hoarseness of voice is a common symptom of :

- ☒ A. Aortic aneurysm
- ☐ B. Aortic dissection
- ☐ C. Coarctation of the aorta
- ☐ D. Aortic regurgitation

37- The best tool for evaluation of posterior cardiac structures

- ☐ A. Transthoracic Echocardiography
- ☐ B. ECG
- ☐ C. Chest X-ray
- ☒ D. Transesophageal echocardiography

38- The left bronchial vein drain in which of the following:

- ☐ A. azygos vein
- ☒ B. hemi-azygos
- ☐ C. Superior vena cava
- ☐ D. Pulmonary vein

39- Which test is NOT used during diagnosis or monitoring of asthma?

- ☒ A. Diffusing capacity.
- ☐ B. Peak expiratory flow rate.
- ☐ C. Bronchial challenge test.
- ☐ D. Skin prick test.

- 40- Which of the following is a cause of localized bronchiectasis?
- A. Recurrent chest infections
 - ☒ B. Foreign body aspiration.
 - C. Cystic fibrosis.
 - D. Immotile cilia syndrome.
- 41- Both immotile cilia syndromes and cystic fibrosis lead to recurrent pneumonia, bronchitis, and eventually bronchiectasis. Which of the following manifestations is typical of Kartagener's Syndrome?
- A. Intestinal obstruction.
 - ☒ B. Dextrocardia.
 - C. Steatorrhea.
 - D. Interstitial pulmonary fibrosis.
- 42- Acute Respiratory Distress Syndrome (ARDS) can be defined as
- A. Sudden life-threatening deterioration of gas exchange in the lungs
 - B. Cardiac pulmonary edema with increasing hypoxemia despite treatment with O₂
 - ☒ C. Sudden life-threatening non cardiogenic pulmonary edema that causes a deterioration of gas exchange despite treatment with O₂
 - D. Gradual life-threatening pulmonary edema associated with cardiomegaly and bilateral pulmonary congestion.
- 43- What is the most common cause of high protein content pleural effusion with normal serum protein?
- ☒ A. TB.
 - B. Liver cirrhosis.
 - C. Urinothorax.
 - D. Heart failure.
- 44- A 45-year-old, female, presented with dyspnea and a dry cough. Chest X-ray showed a left-sided cavity with an air-fluid level, the patient started broad-spectrum antibiotics with no response for 10 days. What is the next step?
- ☒ A. Bronchoscope for microbiological analysis of bronchoalveolar lavage.
 - B. Referral to cardiothoracic surgeon.
 - C. Thoracoscopic biopsy.
 - D. Transthoracic percutaneous needle aspiration (CT- or US-guided).
- 45- Which of the following is the best diagnostic method of mesothelioma?
- A. Pleural fluid chemical analysis.
 - B. Computed tomography scan of the chest.
 - C. Chest x-ray.
 - ☒ D. Histopathological analysis of pleural tissue biopsy.
- 46- Which of the following is highly sensitive for predicting the absence of PE and is used for patients whose probability for PE is unlikely and who may need only minimal additional testing?
- A. V/Q scanning
 - ☒ B. D-Dimer testing
 - C. CT angiography
 - D. Duplex ultrasonography
- 47- A 76-year-old woman is admitted with right pleuritic chest pain and breathlessness. She had surgery 6 months ago for fracture of right femur following a fall. She has a pyrexia of 37°C. Her CXR shows a minimal right basal shadowing. Her serum D-dimers are elevated at 900. White cell count is 14x10⁹/l. What is the most appropriate management plan?
- A. Blood cultures
 - ☒ B. Spiral CT chest
 - C. Start intravenous antibiotics
 - D. Ventilation-perfusion scan.

48- Which of the following is not diagnosed via Spirometry?

- A. Pulmonary fibrosis.
- B. Bronchial asthma.
- C. COPD.
- ☒ D. Pneumonia

49- Which of the following bronchogenic carcinoma is commonly associated with cavitation?

- A. Bronchoalveolar carcinoma.
- B. Large cell carcinoma.
- C. Small cell lung cancer.
- ☒ D. Squamous cell carcinoma.

50- A 40-year- old female who has been nonsmoker and healthy, had a routine chest X- ray and found to have a 3 cm, left upper lobe lung mass, which one of the following bronchogenic carcinoma is most likely to have?

- ☒ A. Adenocarcinoma.
- B. Large cell carcinoma.
- C. Mesothelioma.
- D. Squamous cell carcinoma.

51- Which one of the following symptoms is rare in patients with progressive lung cancer?

- A. Hemoptysis.
- B. Hypercalcemia.
- C. Headache.
- ☒ D. Urinary retention

52- Which of the following drugs can be used as analgesic in case of acute pulmonary edema?

- A. Remifentanyl
- B. Sufentanyl
- ☒ C. meperidine
- D. Diphenoxylate

53- Which of the following is an indication of Direct renin inhibitors ?

- ☒ A. Only hypertension
- B. Resistant heart failure
- C. Acute coronary syndrome
- D. Hypertension with arrhythmia

54- Which of the following is a precaution during treatment by nesiritide ?

- A. Should be combined with beta blockers
- B. Must be used as monotherapy
- ☒ C. ACEIs must be stopped before its administration
- D. ARBs is preferred to be combined with it

BNP

55- Calcification of the periphery of hilar lymph nodes ("egg-shell" calcification) is showed in:

- A. Asbestosis
- B. Sarcoidosis
- ☒ C. Silicosis
- D. Bagasosis

56- As a risk reduction strategy, which one of the following vaccination should be recommended to all chronic obstructive pulmonary disease patients beside a yearly influenza vaccination ?

- A. Herpes zoster vaccine
- ☒ B. Pneumococcal vaccination
- C. Haemophilus influenza vaccine
- D. Rubella vaccination

57- Typical bacterial pneumonia is caused by:

- ☒ A. Staphylococcus aureus.
- B. Legionnaires pneumonia.
- C. Pneumocystis carinii.
- D. Mycoplasma pneumoniae.

58- Which of the following would not be an expected sign of right-sided congestive heart failure?

- A- Prominent jugular vein
- B- Hepatomegaly
- C- Bilateral pitting lower limb edema
- ☒ D- Bilateral fine basal crepitation

59- The following is recognized cause of dilated cardiomyopathy:

- A. Systemic sclerosis.
- B. Polymyositis.
- ☒ C. Hemochromatosis.
- D. SLE.

60- A 2-month-old girl presents to the emergency department with respiratory distress. Her mother reports that the infant had been well until about 2 weeks ago when she first noticed the infant had rapid breathing.. The infant had previously fed well but now becomes sweaty and irritable when taking a bottle. The mother is concerned that the infant is losing weight. Physical examination reveals a HR of 140/min, BP 80/45 mmHg, RR 60/min, and temperature 36.8 °C. There is a prominence of the left precordium and a palpable sternal lift. She has a blowing holosystolic murmur that is best heard along the left sternal border. SpO2 is 95%. Her chest x-ray shows cardiomegaly and increased pulmonary vascular markings. Which of the following is the most likely explanation for this child's symptoms?

- A. Right-to-left shunt
- ☒ B. Left-to-right shunt
- C. Left ventricular outflow tract obstruction
- D. Right ventricular outflow tract obstruction

2) SAQ

- 1) Enumerate 4 causes of hypoxemic respiratory failure. (1 mark)

pneumonia
Acute Respiratory distress syndrome (ARDS)
Pulmonary embolism
pulmonary edema

- 2) Enumerate 4 causes of pneumothorax. (1 mark).

chest injury rib fracture
Lung disease COPD
Ruptured air blisters
mechanical ventilation

- 3) Enumerate two causes of irregular narrow complex tachycardia (1 mark)

Atrial Fibrillation
~~Atrial~~ Atrial flutter

- 4) List two complication of acute myocardial infarction (1 mark).

Heart Failure
Arrhythmias

- 5) List 4 structures of the posterior relation of the heart? (1 mark).

Left atrium
Esophagus
Descending aorta
Thoracic vertebra

- 6) Enumerate two therapeutic uses of phosphodiesterase enzyme inhibitors in respiratory diseases? (1 mark)

treatment of chronic obstructive pulmonary disease COPD
management of pulmonary hypertension

7) Enumerate 4 causes of secondary hypertension (1 mark).

...chronic...kidney...disease.....
 ...primary...aldosteronism.....
 ...pheochromocytoma.....
 ...Cushing's...syndrome.....

8) List four factors affecting pulmonary blood pressure and flow: (1 mark).

...cardiac...output.....
 ...Pulmonary...Vascular...resistance.....
 ...Blood...Volume.....
 ...Pulmonary...artery...pressure.....

3) EMQ

- 9) Each group of extended matching questions consists of a list of numbered problems (diseases) followed by lettered options (clinical symptoms). For each numbered disease select the lettered symptoms that most closely diagnose the disease. You can use the lettered options once or not at all.

Total 2 marks (0.5 mark for each question)

1. Problems:

1. Pulmonary embolism. **G**
2. Pleural effusion. **A**
3. Lung abscess. **B**
4. Mediastinal tumor. **D**

Options:

- A. Pleuritic stitching pain followed by gradual dyspnea.
- B. Fever and productive cough of 150 cc sputum per day.
- C. Productive cough, dyspnea, and chest wheezes.
- D. Brassy cough, dysphagia, and hoarseness of voice
- E. Dry cough, chest wheeze and chest pain.
- F. Dyspnea, orthopnea, and paroxysmal nocturnal dyspnea.
- G. Acute dyspnea, pleuritic chest pain, hemoptysis, and cyanosis.

10) Select from column A the correct answer from column B (2-mark 0.5 mark each)

Column A	Column B
1-Acute coronary syndrome C	A- Stitching chest pain relived by leaning forward.
2- Acute Aortic dissection D	B- Localized inframammary staping chest pain
3- Acute pericarditis A	C- Compressing retrosternal chest pain referred to left shoulder.
4-Cardiac neurosis B	D- Tearing chest pain radiated to the back .

4) MEQ

11) Give short account on stap wound at left 4th intercostal space (2 marks)

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

12) Describe the treatment of acute severe asthma.(2 marks)

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

5) CASES

13) Case 1

(2 marks)

A 60-year-old male patient, smoker, diabetic, and lives in poorly ventilated house. He attended outpatient's clinic suffering from night fever, night sweating, hemoptysis, anorexia, and loss of weight, with high ESR and chest x-ray revealed cavity with nodular infiltration in right apex.

A. What is the most probable diagnosis? (0.5 mark)

.....
.....
.....
.....

B. Enumerate 2 risk factor in this patient. (0.5 mark)

.....
.....
.....
.....

C. What are the 2 investigations you advise to clinch the diagnosis? (1 mark)

.....
.....
.....
.....
.....

14) Case 2

(2 marks)

A 60-year-old male, smoker, farmer, and bird breeder. He presents with six months history of worsening breathlessness and dry cough, Examination revealed central cyanosis, 2nd-degree clubbing, with fine, late inspiratory, and bi-basal crackles all over the lung field. Chest radiography shows bilateral reticulonodular infiltration. His spirometry test revealed low FVC with normal FEV1% and FEV1/FVC ratio.

A. What is the most likely diagnosis? (0.5 mark)

.....
.....
.....

B. Mention the 2 diagnostic tests for this disease (0.5 mark)

.....
.....
.....

C. List 2 medications used to treat this disease. (1 mark)

.....
.....
.....
.....

15) Case 3

(6 marks)

30 year old IV drug user presents with shortness of breath, cough , night sweats and a fever. On examination, you find that he has splinter hemorrhages.

Vital signs:

- Pulse: 100 bpm, regular and equal
- BP: 115/ 68 mmHg equal in both side
- Respiratory rate: 35 c/m with O2 sat. 90%
- Chest auscultation: coarse rhonchi and absent breath sounds
- Cardiac auscultation: normal S1, S2 with pan systolic murmur at lower left sternal border increased with inspiration .

X ray chest : irregular shaped cavity with an air fluid level (lung abscess)

A. What is the most likely diagnosis? (2mark)

.....
.....

B. Mention two investigation needed to confirm diagnosis? (2mark)

.....
.....
.....

C. List two other peripheral sign may be present on this patient? (2mark)

.....
.....
.....

16) Case 4

(2 marks)

A 40-year-old male, Mr.S, presented to out patient clinic with difficulty breathing on stair climbing to his office on 2nd floor for almost a year. He complains of persistent cough with expectoration for the last 6 years, which increases every 2-3 months & then goes away for some time. He was a cigarettes smoker for twenty year consuming about 1 1/2 packs per day.

By examination: he has moderate respiratory distress & tachypnea.

Lung examination reveal bilateral equal diminished air entry & expiratory wheezes

Chest X-ray revealed: widening of intercoastal spaces, flattened diaphragm & increased bronchovascular markings

What is the most likely diagnosis of this case?

.....
.....

17) Case 5

(2 marks)

Ali is a 4-year-old boy who is admitted to the paediatric ward from the emergency department (ED) with pneumonia. He has had many courses of oral antibiotics over the past few years from his general practitioner (GP) for chest infections. He also has recurrent abdominal pain, and his parents report large offensive stools. He is small (height 9th centile and weight 2nd centile), pale-looking, and miserable. He has finger clubbing. His temperature is 38.7°C, respiratory rate 40 breaths/min, heart rate 140 beats/min and oxygen saturation 89 per cent in air. There is reduced air entry at the left base with bronchial breath sounds in the left midzone, and coarse crackles are heard on both sides of the chest. Cardiovascular examination is unremarkable. His abdomen is mildly distended but non-tender.

What is the likely underlying diagnosis? (1 mark)

.....

.....

What investigation would you do to confirm the diagnosis? (1 mark)

.....

.....

GOOD LUCK